v1-mini Simulation & Performance Report

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Date	April 14th, 2026
Document Version	v1.0

This document provides a summary of the RG Sonic32 v1-mini simulation, including functional validation and cycle-level execution metrics.

Note: *This simulation report was finalized after the hardware implementation report. Minor details may differ due to subsequent design updates.*

Overview

This document summarizes the RTL simulation results for the RG Sonic32 v1-mini CPU. Validation was performed using a cycle-accurate testbench to verify correct instruction execution, register writeback behavior, branch handling, and successful program completion. In addition to functional correctness, the simulation provides cycle-level visibility into execution flow, enabling basic runtime performance characterization. All validation checks passed, with correct final register states observed and successful program termination within the expected cycle count.

Testing Configuration

Item	**Value**
Testbench	tb_cpu_top.v
Simulation Type	RTL (cycle-accurate)
Timescale	1 ns / 1 ps
Clock Period	10 ns (100 MHz)
Reset Duration	30 ns (active low)
Memory Size	65,536 bytes
Program Image	test_mem.hex
Termination Condition	HALT signal or 500-cycle timeout

Results & Execution Trace

Functional Results

All architectural state checks passed at the conclusion of simulation. Final register values matched expected results, and the processor successfully reached the halt condition without triggering the timeout.

Register	Expected	Observed	Status
r4	10	10	PASS
r5	20	20	PASS
r6	30	30	PASS
r7	256	256	PASS
r8	30	30	PASS

Halt Signal: Asserted
Timeout: Not triggered
Simulation Status: ***ALL CHECKS PASSED (yay)***

Condensed Execution Trace

Key execution events extracted from the simulation log are shown below:

```
[cyc 7]  WB IR=0480000a -> r4 = 10
[cyc 13] WB IR=04a00014 -> r5 = 20
[cyc 19] WB IR=00c42800 -> r6 = 30
[cyc 25] WB IR=04e00100 -> r7 = 256
[cyc 42] WB IR=29070000 -> r8 = 30
[cyc 52] BRANCH Z=1 N=0 -> branch taken
[cyc 57] HALT reached
```

The trace confirms correct instruction sequencing, register writeback behavior, and branch resolution. Repeated FETCH cycles between events are consistent with the multi-cycle execution model of the processor.

Performance Metrics

Cycle-accurate RTL simulation was used to extract execution and performance characteristics of the v1-mini CPU. The test program completed in 57 cycles with a total of 9 instructions executed.

Core Metrics

Metric	Value
Total Cycles	57
Instructions Executed	9
Average CPI	6.3
Clock Period (simulation)	10 ns
Estimated Clock (Artix-7)	~80 MHz
Estimated Throughput	~12.7 MIPS at estimated hardware clock

Instruction-Level Latency

Instruction Type	Cycles
ADD / ADDI	6
SUB	6
Store (SW)	8
Load (LW)	9
Branch (BEQ)	5

HALT	1

Performance Analysis

- The processor exhibits an average CPI of 6.3, consistent with a multi-cycle FSM-based architecture.
- Memory operations dominate execution time, with loads requiring the longest latency (9 cycles) due to memory access and writeback.
- Arithmetic instructions complete in 6 cycles, reflecting datapath reuse across sequential stages.
- Branch instructions introduce a control penalty, requiring ~5 cycles including resolution.
- At an estimated 80 MHz operating frequency, the design achieves approximately 12.7 MIPS, aligning with expected performance for a non-pipelined CPU.

Performance Interpretation

These metrics reflect cycle-level behavior derived from RTL simulation and serve as a baseline characterization of the v1-mini architecture. Future versions (v2) will introduce pipelining to significantly reduce CPI and increase overall throughput.

Architectural Observations

- The processor operates as a multi-cycle finite state machine (FSM), with instructions progressing through sequential stages rather than overlapping execution.
- Repeated FETCH cycles between operations confirm a staged control flow, where datapath resources are reused across cycles to minimize hardware complexity.
- Instruction completion is marked by discrete writeback events, indicating deterministic execution timing for each instruction type.
- Memory operations introduce the highest latency, reflecting the additional cycles required for memory access and data transfer within the multi-cycle design.
- Branch instructions incur a control penalty, as execution must evaluate condition flags before updating the program counter, resulting in visible control flow delays.

- The observed execution behavior aligns with expectations for a non-pipelined architecture, trading higher CPI for reduced design complexity and improved control clarity.

Next Steps & Limitations

Limitations

- Internal processor state is currently not observable on hardware, as v1-mini does not include a UART/debug interface.
- Detailed execution traces and performance metrics are derived from RTL simulation only.
- Performance estimates (frequency and MIPS) are based on pre-synthesis assumptions and not yet validated on hardware.

Next Steps

- Integrate a UART-based debug module to enable real-time register and state visibility on hardware.
- Perform full hardware validation, including execution trace verification and runtime behavior.
- Extend performance analysis with measured hardware timing and throughput.
- Begin architectural enhancements for v2, including pipelining to reduce CPI and improve overall performance.